

“ TO **AVOID** TEDIOUS REPETITION OF... ‘**IS EQUAL TO**’, I WILL SETTLE... ON A **PAIR OF PARALLELS** OF ONE LENGTH... ”

Robert Recorde, Welsh physician and mathematician, 1557

Robert Recorde was the author of the first books on algebra in English.



Smoking became a fashionable habit in 16th-century Europe.



#### **De Re Metallica**

Georgius Agricola's lavishly illustrated book on mining techniques describes the formation of ores in the ground, and how metal can be extracted from them.

descriptions of the veins of ores found in rock, and how metals could be extracted from them, as well as a comprehensive catalog of the minerals known at that time. Sometimes referred to as “the father of mineralogy,” Agricola made contributions to the emerging fields of geology, metallurgy, and chemistry.

After his books *The Ground of Artes* (1543) on arithmetic, and *The Pathway to Knowledge* (1551) on geometry, the **Welsh mathematician**

**GEORGIUS AGRICOLA'S BOOK** *De Re Metallica* (*On the Nature of Metals*) was published posthumously in 1556. In it, he described various techniques for mining minerals, and the machines, especially water mills, used for raising them from the mines. This **classic text of mining engineering** included

Robert Recorde published a companion volume, *The Whetstone of Witte*, in 1557, probably the **first book on algebra in English**. As well as presenting the principles of algebra, this book established usage of the symbols + (plus) and – (minus), which had previously been used only occasionally by

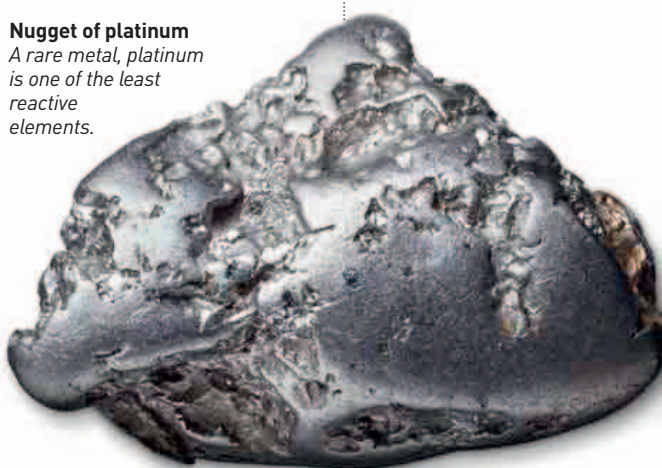
some German mathematicians. It introduced a symbol he invented: =, **the equals sign**. Best known for popularizing mathematics in Britain, Recorde had originally studied medicine and worked as a physician to the Royal family, and for a time was

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## THE MELTING POINT OF PLATINUM

controller of the Royal Mint supervising the manufacture of coins. Despite fame and standing, he died in a debtors' prison a year after publishing *The Whetstone*.

**Nugget of platinum**  
A rare metal, platinum is one of the least reactive elements.



Although **platinum** had already been used by the indigenous people of Central and South America to make jewelry and ornaments, it was unknown in Europe until the 16th century. The **first written reference** to the metal came in 1557 in the writings of **Julius Caesar Scaliger** (1484–1558), an Italian scholar. He described how Spanish explorers came across an unknown element, with an **unusually high melting point and resistance to corrosion**. Originally known as “white gold,” platinum was later recognized as an element which occurs naturally in both pure and alloy (combined with another element) forms in South America, Russia, and South Africa.

**FRENCH DIPLOMAT JEAN NICOT**, (1530–1600) while ambassador in Lisbon, Portugal, was introduced to **tobacco, brought by Spanish explorers from America**. Native Americans smoked it in religious rituals, and ingested or made poultices with its leaves for medicinal purposes. Nicot sent tobacco plants and snuff to the **royal court in France**, where smoking and snuff-taking soon became fashionable. The tobacco plant *Nicotiana*, and the chemical nicotine, are named after him.

In the same year, Italian anatomist and surgeon **Realdo Colombo** (c.1516–59) published



**Title page of De Re Anatomica**  
Although Realdo Colombo published only one book on anatomy, his discoveries rivaled those of Andreas Vesalius and Gabriele Falloppio.

1556 Georg Pauer's *De Re Metallica* is published in Basel under the Latinized form of his name, Georgius Agricola

1557 First European reference to platinum in writings of Julius Caesar Scaliger

1557 Robert Recorde introduces the symbol =, for “equals,” in *The Whetstone of Witte*